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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2024

CC3214-1- WIRELESS SENSOR NETWORKS
(For CC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is the role of the Guaranteed Time Slot (GTS) in IEEE 802.15.4?**
A) To allocate dedicated time slots for devices to transmit or receive data without contention
B) To ensure fixed-length frames are transmitted
C) To prevent all devices from sleeping simultaneously
D) To provide additional energy-saving modes
2. **In IEEE 802.15.4, which type of device acts as PAN coordinator?**
A) Reduced Function Device (RFD)
B) Both FFD and RFD
C) Full Function Device (FFD)
D) Only devices with ZigBee support
3. **Which of the following features is unique to TRAMA compared to S-MAC?**
A) Use of RTS/CTS handshake
B) Two-channel wakeup system
C) Distributed scheduling of collision-free time slots
D) Periodic wakeup scheme
4. **In TRAMA, how is the transmission schedule determined?**
A) Using a global hash function to compute priority values
B) Through random access periods
C) By polling the central coordinator
D) Based on a node's residual energy
5. **In the RTS/CTS mechanism, what does another node do when it hears an RTS or CTS packet?**
A) It sends a busy-tone signal to the network
B) It starts its own transmission immediately
C) It sets a Network Allocation Vector (NAV) and waits
D) It acknowledges the RTS packet
6. **How does Ultra Wide Band (UWB) communication transmit digital signals?**
A) By modulating them onto a specific carrier frequency
B) By using frequency shift keying
C) By utilizing amplitude shift keying
D) By transmitting very short impulses across a large bandwidth
7. **What is the primary advantage of a wakeup radio system in wireless communication?**
A) It allows receivers to stay powered off until a packet is detected.
B) It increases the data transmission speed
C) It enhances the signal strength for better transmission.
D) It reduces the transmission error rates.
8. **Which of the following is NOT a feature of radio frequency (RF) communication?**
A) Long range and high data rates
B) Acceptable error rates and energy
C) Requires line of sight
D) Limited to 433 MHz to 2.4 GHz

9. What limits a packet's lifespan in flooding?
 - A) Hop count or TTL
 - B) Node energy levels
 - C) Bandwidth
 - D) Data packet
10. What happens in traffic implosion during flooding?
 - A) Duplicate packets at a node.
 - B) Packets are dropped due to congestion.
 - C) Nodes stop forwarding packets.
 - D) Energy consumption is reduced.
11. What message type indicates interest in SPIN data?
 - A) ADV (Advertisement)
 - B) REQ (Request)
 - C) DATA
 - D) ACK (Acknowledgment)
12. What is a key goal of energy-aware routing in WSNs?
 - A) Maximize network lifetime
 - B) Minimize data packet loss
 - C) Maximize data transmission speed
 - D) Reduce network complexity
13. How are probabilities assigned to neighbors in routing?
 - A) Based on distance only
 - B) Inverse to energy cost
 - C) Based on node energy levels
 - D) Equal probability for all
14. What does infrequent flooding in route maintenance achieve?
 - A) Keeps paths alive
 - B) Increased energy consumption
 - C) Reduced data accuracy
 - D) Creates network
15. What type of network is Rumor Routing suitable for?
 - A) Low event, high query
 - B) High bandwidth networks
 - C) Networks with static nodes
 - D) Networks with centralized control
16. What does GBR assign to each node in the network?
 - A) A height
 - B) A data aggregation role
 - C) A unique identifier
 - D) An energy level threshold
17. Which approach does TinyOS use to manage tasks?
 - A) Preemptive multithreading
 - B) Multicore processing
 - C) Event-driven model
 - D) Batch processing
18. Which operating system is designed to ensure concurrent data flows among hardware devices?
 - A) Windows
 - B) Linux
 - C) TinyOS
 - D) MacOS
19. What is the primary method of inter processor communication in EMERALDS?
 - A) Shared memory
 - B) Direct disk access
 - C) External APIs
 - D) File transfers
20. Which sensor node type is most suitable for acting as a gateway to the internet?
 - A) Spec node
 - B) Berkeley motes
 - C) iMote
 - D) Stargate

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Point out the characteristics and major challenge of wireless sensor networks.	8	L*2	1	1
	b) Elaborate the difference between STEM-B and STEM-T.	4	L2	2	1
	c) Discuss the protocol that provides mechanisms to circumvent idle listening, collisions, and overhearing.	4	L2	2	1
2.	a) With examples discuss the mechanisms that form parts of WSNs.	8	L2	1	1
	b) Compare the performance of TRAMA with S-MAC	4	L2	2	1
	c) Explain the protocol that allowing nodes to access a single channel in a collision-free manner.	4	L2	2	1
3.	a) Discuss the main sensor node hardware components with suitable block diagram.	8	L2	1	1
	b) Illustrate the wakeup concepts for MAC protocols with timing diagram.	4	L2	2	1

- c) List out the different types of energy scavenging approaches. Explain any two.

4 L2 1

Unit – II

4. a) Discuss the strategy to balance the competing needs of the routing challenge.
 b) What are main objectives of Low-Energy Adaptive Clustering Hierarchy (LEACH) in WSNs. How to achieve the objectives of LEACH
 c) Illustrate the chain-based approach in WSNs with suitable example.
5. a) Describe the different class to address the design requirements, routing strategies for WSNs.
 b) What are the main objectives of Sensor Protocols for Information via Negotiation? Explain it with example of node diagram.
 c) Explain the working of Threshold sensitive Energy Efficient Sensor Network Protocol with suitable diagram
6. a) Discuss the protocol for information gathering and dissemination in WSNs. Point out its limitations and advantages
 b) Discuss the working and importance of Greedy Perimeter Stateless Routing (GPSR) in WSNs.
 c) With example describe the principal mechanics of data aggregation. Point out its benefits.

8 L2 3 1
 4 L2 4 1
 4 L2 4 1
 4 L2 3 1
 4 L2 3 1
 8 L2 1
 8 L2 3 1
 4 L2 4 1
 4 L2 4 1

Unit – III

7. a) Point out the sensor nodes groups with examples and its advantages over traditional OS.
 b) Point out the difference between MagnetOS and MANTIS Operating Systems for Wireless Sensor Network.
8. a) Discuss the Operating System Design Issues and Sensor operating systems (SOS) functions in WSNs.
 b) Explain the following terms. (i) OSPM and (ii) EYES OS

8 L2 5 1
 8 L2 5 1
 8 L2 5 1
 8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
